Sensor Calibration Program

Calibration Software for Wind Tunnel System 8421

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Technical Data are subject to change!

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INTRODUCTION

The Sensor Calibration Program has been developed for calibration of wind speed sensors in Wind Tunnel, type 8421 and compatible.

The complete system consists of

1. Wind Tunnel 8421.0000, open construction with measurement section in suction part, incl. 19" drive and control cabinet.
2. Flow Control Box (**FCB**) contenting the following parts:
    a. Thermal Anemometer as reference device; velocity range 0…7,5 m/s.
    b. Micromanometer system for differential pressure measurement; measuring range 0…16 hPa, including a Prandtl tube.
    c. Sensors for air temperature, relative humidity and air pressure to calculate air density.
3. Two Data logger COMBILOG 1022 to measure all sensor values, build in FCB, connected to the PC via RS232.
4. Windows 10 based computer system, including ink jet printer and a PEXDA PC-card for automatic sensor test procedure.
5. A set of data cables.

The program registers the following measuring data from the data loggers COMBILOG 1022 via RS232:

1. Sensor on test with frequency, voltage or current output
2. Wind velocity from Thermal anemometer
3. Differential pressure from Micromanometer system
4. Air temperature
5. Relative humidity
6. Air pressure

There are *2 references* for the *wind velocity* inside the wind tunnel. The *first reference* is the wind velocity which is measured with a Thermal anemometer and the *second reference* is the wind velocity which is calculated with the measurements from the Differential pressure from Micromanometer system, the Air temperature, the Relative humidity and the Air pressure. The *first reference* is suitable for measurements of Threshold up to 6 m/s and the *second reference* is suitable for measurements from 2 m/s and higher.
The output signal of the Differential pressure from Micromanometer system is measured as current. Depending on the range of the devices this current will be calculated to wind velocity, considering the air density, which is calculated from temperature, humidity and air pressure.

Depending on the sensor type that should be calibrated, the user can enter the sensors range limits for wind velocity (m/s or kn) and physical signal output (Hz, V or mA) at the beginning of the calibration process. Then a factor and offset of the sensor will be calculated automatically which allows converting the physical sensor output (Hz, V or mA) to wind velocity values (m/s or kn). Depending on the size of the test sensor, an additional correction factor can be entered to correct the influence of the sensor to the airstream in the tunnel.

Sensors with an output at a serial RS485-interface can be used for reading the wind velocity out of NMEA data telegrams.

Manual input of sensor velocity is also possible.

The program provides the possibility to measure the starting threshold of the sensor under test and the normal measurement of the wind velocity.

The measurements are available as well in automated Test series mode as in One-Point-Measurement mode.

The results will be displayed in alphanumerical and graphical form and can be saved as a file and printed out too.

1 OPERATION

Before starting the operation of the Wind Tunnel system: The operator should be acquainted with the technical descriptions of the different parts of the system. Special attention shall be paid to the layout and function of the push buttons and switches at the front of the control rack.

1.1 PREPARATION OF MEASUREMENT

1. Set main switch “ON”. This will power on the drive and control logic of the Wind Tunnel. If the FCB (Flow Control Box) is already switched ON this will power on the data logger COMBILOG 1022, the Thermal Anemometer and the Micromanometer system. The operating of the data logger will be indicated by the green **RUN** LED. The PC will be powered on as well.
2. Press the “Reset emergency stop” button.
   The blinking “Alarm” lamp shall be off
3. Switch the control button at the front of the 19" cabinet to manual (position 1) or automatic (position 2). The default position of the control button is automatic. If the control button is in position manual, this will let you manually regulate the rotational speed of the Wind Tunnel. If the control button is in position automatic, this will let you regulate the speed of the Wind Tunnel by the software.
4. Switch on the PC system. The Windows 10 operating system is configured for automatic login. Run the Windtunnel.exe program on the PC by clicking the program icon on the desktop or from start menu (Start | All Apps | Windtunnel | Windtunnel.exe).

Figure 1: Main Window at software start

The program firstly connects with the data loggers via RS232 and then takes the values of the Thermal anemometer, Differential pressure from Micromanometer system, the Air temperature, the Relative humidity and the Air pressure via the data loggers automatically.
The values of the ***Thermal anemometer*** which is the *first reference*, the ***Micromanometer*** which is the *second reference*, the ***Air temperature***, the ***Relative humidity*** and the ***Air pressure*** are shown at the right side of the screen permanently.

If the connection could not be established, the message "No answer from COMBILOG" will appear and the program is in offline mode. In this case you should make sure, that the data logger is working and the communication cable is connected correctly.

You can try to reconnect by pressing the 

key on the data panel (only visible in offline mode).

The data will be updated at least one time per second, and the data logger indicates the communication by the blinking **RUN** LED.

In case of implausible or missing data **Error** will be displayed on the data panel.

There are the following two ways to switch ***on*** and ***off*** the control cabinet:

1. Hardware button:

   Switch on the control cabinet by pushing the “ON”-button.
   Switch off the control cabinet by pushing the “OFF”-button.

2. Software button:

   Switch on the control cabinet by clicking with the mouse on the “ON”-button in the main-window of the calibration software.
   Switch off the control cabinet by clicking with the mouse on the “OFF”-button in the main-window of the calibration software.

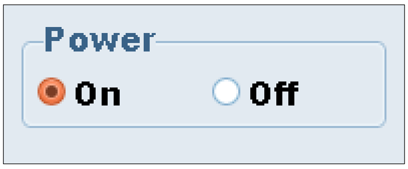

Figure 2: Software button

1.2 MEASURING PROCEDURE

1.2.1 SELECTION OF SENSOR TYPE

Choose File | New… to create a new measuring protocol or click to .

In the following dialog box you can specify a sensor, that should be calibrated:

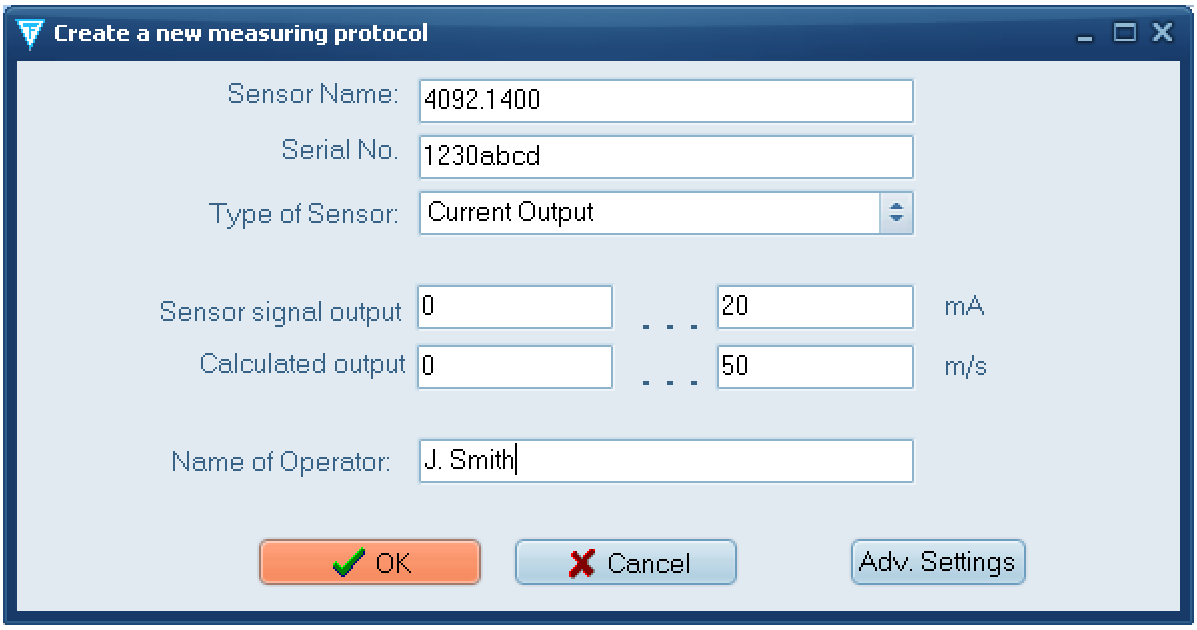

Figure 3: Example of creating a new measuring protocol

Sensor Name: any text (max. 30 character) to describe the sensor, e.g. manufacturer type name

Serial No.: any text (max. 20 character) for sensor identification

Type of Sensor: selection of the output signal of the sensor to determine, at which channel the data logger COMBILOG 1022 measures the sensor signal

Frequency Output: The sensor provides a digital pulse output, that is measured as frequency. Signal levels between 3,5V and 18V are interpreted as logic LOW, voltages lower than 1V as logic HIGH

Voltage Output: The sensor provides an analogue voltage corresponding to the wind velocity. Possible range: ±10VDC

Current Output: The sensor provides an analogue current corresponding to the wind velocity. Possible range: 0…20mA

NMEA Output: The sensor provides an ASCII string output, that is connected via a serial RS485-Interface to the PC.
Possible protocol: NMEA V2.0.

Manual Measurement: In this case it is not possible to measure the sensor signal by one of the above methods, it must be measured by external methods. In this case the sensor value can be entered manually.

Sensor Signal Output:

Selection of the output signal range of the sensor.

Calculated Output: Selection of the calculated signal range of the sensor.

The *Sensor Calibration Factor* and the *Offset* are automatically calculated by selecting the ***Sensor Signal Output*** and ***Calculated Output*** and must not be entered separately.

Sensor Calibration Factor:

In case of selection of sensor type with frequency, voltage or current output a calibration factor must be known to calculate the wind velocity from frequency, voltage or current. The conversion formula is:

velocity [m/s] = measured value [Hz, V or mA] * Factor + Offset [m/s]

Example:

A sensor of the type 4092.1400 has a signal output as current: 4..20mA means 0..50 m/s. Therefore the factor is:

50 [m/s] / 16 [mA] = **3.125** m/s per mA.

Offset: This is the offset in m/s in the above formula

Example:

The sensor offset in the above example can be calculated as follows:
We know that at a measured value of 20 mA the velocity is ideally 50 m/s. Therefore the offset is:

50 [m/s] - 20 [mA] * 3.125 [m/s / mA] = 50 [m/s] – 62.5 [m/s] = **-12.5** [m/s].

Operator: any text (max. 20 character) to enter the name of the operator

: Pressing this Button opens a new Window with advanced settings, which are described in the following subchapter.

1.2.1.1 ADVANCED SETTINGS

Depending on which Type of Sensor is chosen, you can determine additional specifications of the Sensors:

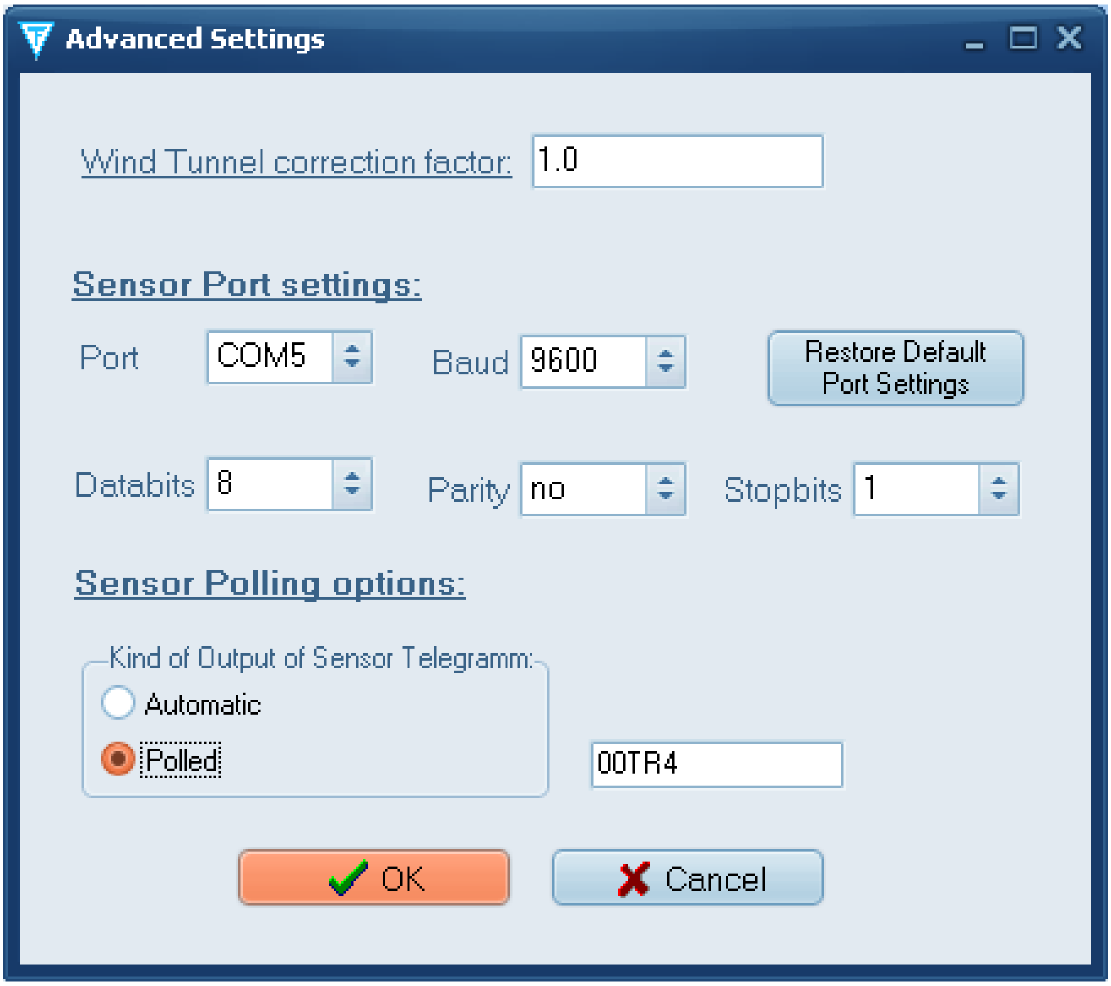

Figure 4: Advanced Settings

Wind Tunnel correction factor:

This is an additional factor to correct the influence of the sensor to the airstream in the Wind Tunnel in case the sensor is to big for the measuring section. This factor is to find out empirically for each size of sensor. Set this value to 1.0 in case there is no specific factor available.
The sensor velocity v will be calculated by the following formula:

$$\boxed{v = \text{measured value} * \text{SensorCalibrationFactor} * \text{CorrectionFactor} + \text{Offset}}$$

Please note, that this is only an additional add-on without the given accuracy from the specifications.

Sensor Port settings:

This settings are just displayed for the Type of Sensors with NMEA output. *You shouldn't change these settings if it isn't necessary.* Between the RS485-Interface of the NMEA Sensor and the RS232-interface of the PC there is a RS485-to-RS232-converter.

Port, Baud, Databits, Parity and ***Stopbits*** specifies the serial RS232-interface of the PC to which the ASCII-strings of the NMEA Sensors are sended.
For resetting the Sensor Port Settings to its default, please press the

button.

Sensor Polling options:

This options are just displayed for the Type of Sensors with NMEA output.
You shouldn't change these settings if it isn't necessary.

By default the NMEA sensor is a polled sensor, that is the data string must be requested by a Request Command, which is '**00TR4**'. If other is needed you can edit the request command right to the box 'Kind of output of sensor telegram'.

If the sensor sends the data string automatically choose *Automatic*, that is no Request Command is needed. In this case the sensor should send the data string at least every second.

1.2.2 STRUCTURE OF A MEASURING PROTOCOL

After creating a new measurement an empty measuring protocol is shown automatically, containing the sensor description and a table for measuring values.

A protocol can also be opened from file by ***File | Open*** , to display the results of an earlier measurement.

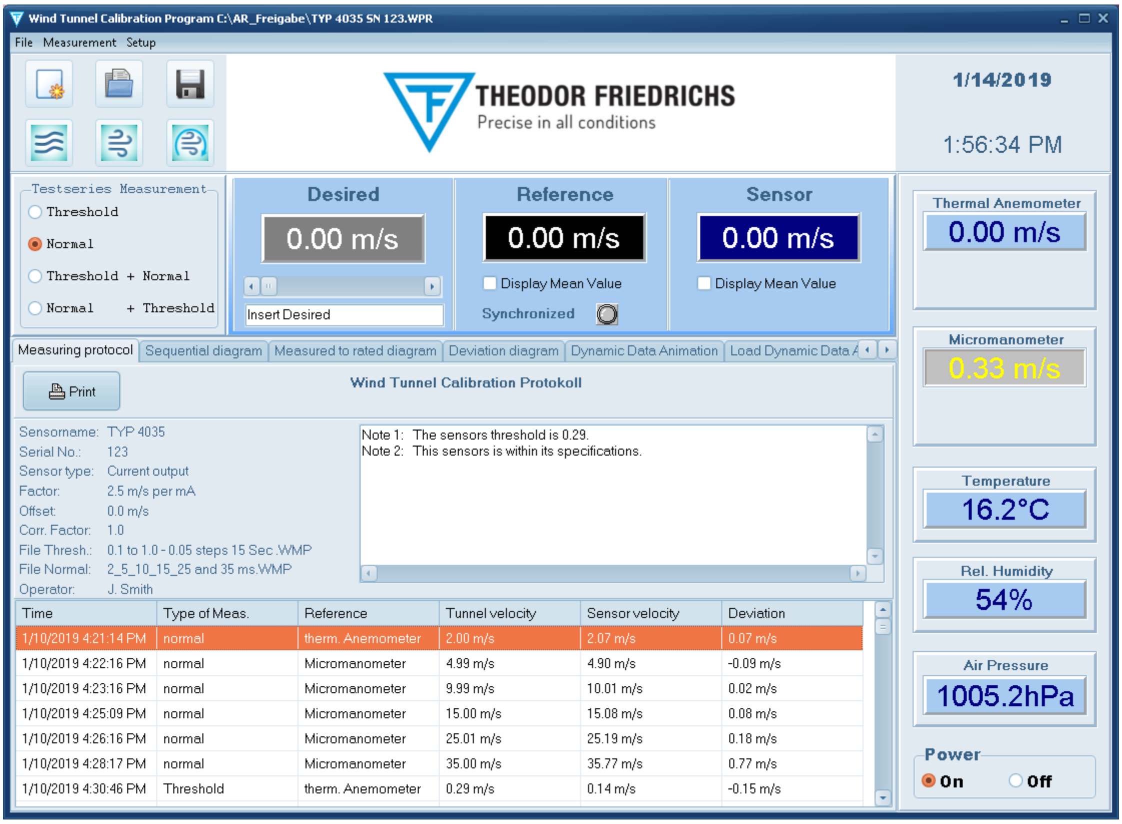

Figure 5: Main Window with a Measuring protocol

The measuring table contains the following information:

Time:	date and time of the performed measurement
Type of meas.:	characterises the performed kind of measurement. Possible values are ***Threshold*** for measurement of starting threshold and ***normal*** for velocity measurement.
Reference:	characterises the actual used reference device for measurement of the tunnel velocity: ***therm. Anemometer*** in case the thermal anemometer is used, or ***Micromanometer*** in case of the micromanometer
Tunnel velocity:	value of the tunnel velocity
Sensor velocity:	value of the sensor velocity
Deviation:	difference between sensor and tunnel velocity

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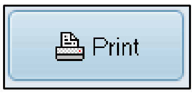
All measuring values are displayed in the predefined unit (see program settings). In case the unit will be changed, all values will be recalculated.

The measuring protocol can be send to a printer by clicking the 

button.

To save the measuring results to file click to  and enter a file name. The file extension WMP will be appended automatically.

To make the measuring data available for other programs, the data can be stored as ASCII file. Choose File | save as and select the file type text file (*.csv). This file format can be imported directly e.g. to MS Excel.

The measuring protocol consists of 3 additional sheets to display the data graphically in the following diagrams:

Sequential diagram:

Shows the measuring values tunnel velocity and sensor velocity in the order of their recording:

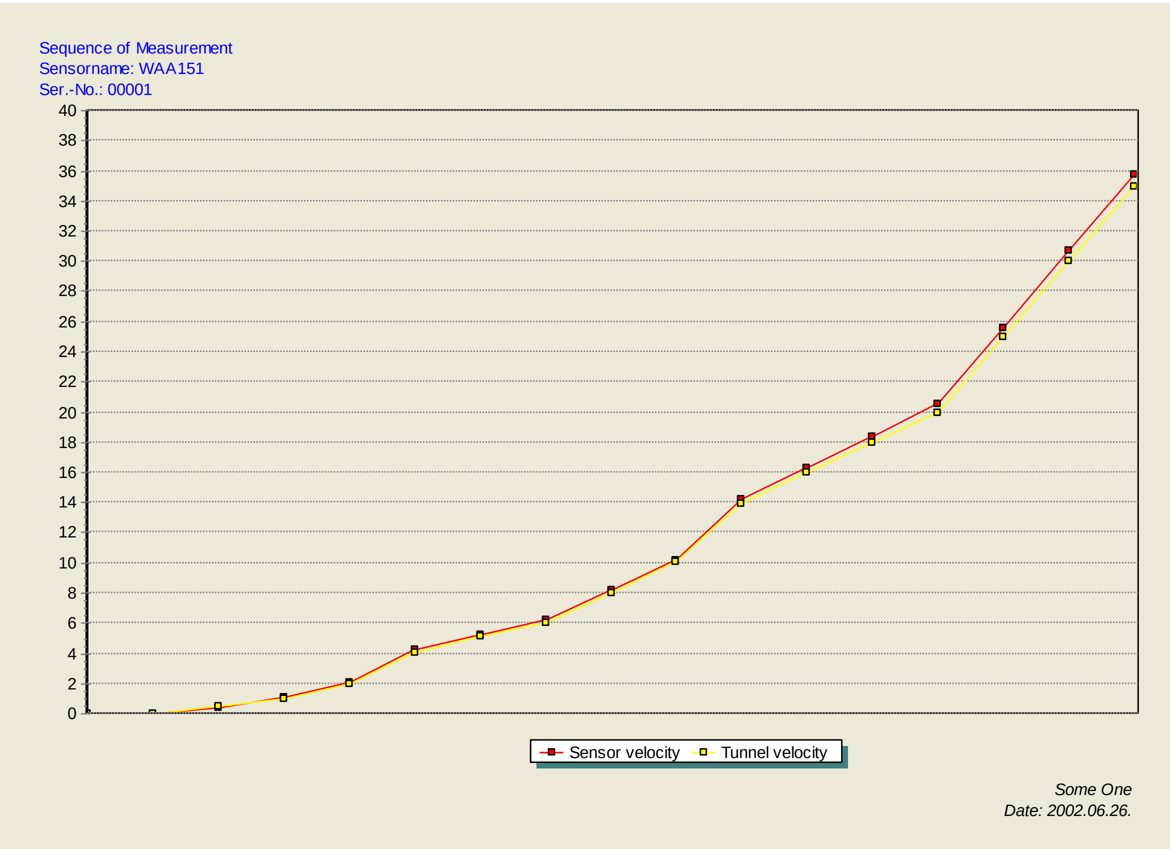


Figure 6: Sequential diagram

The x-axis of this diagram is divided in equidistant steps.

Measured to rated diagram

This diagram shows the sensor velocity (at y-axis) related to the tunnel velocity (at x-axis), so the nonlinearity of the sensor can be seen:

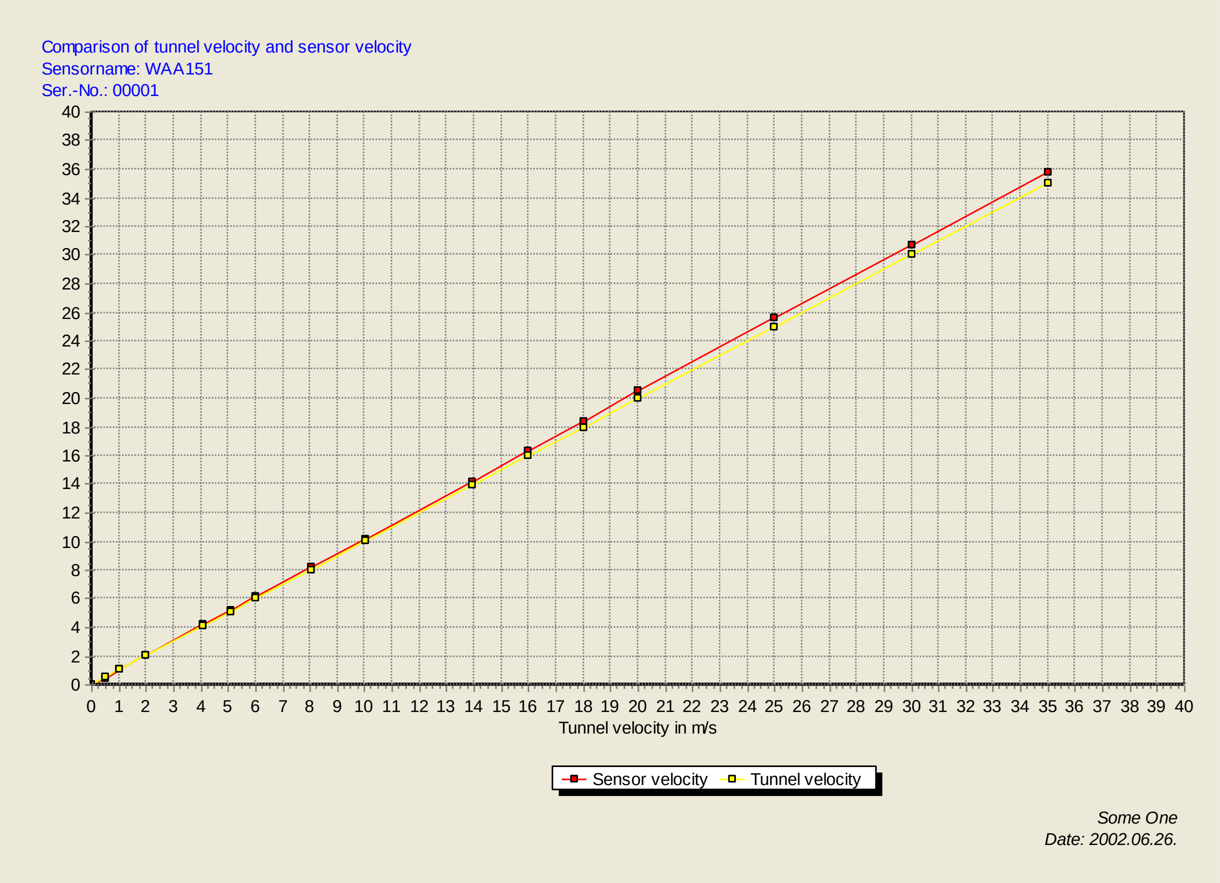


Figure 7: Measured to rated diagram

Deviation diagram

This diagram shows the deviation of the sensor velocity from the tunnel velocity (at y-axis), related to the tunnel velocity (at x-axis):

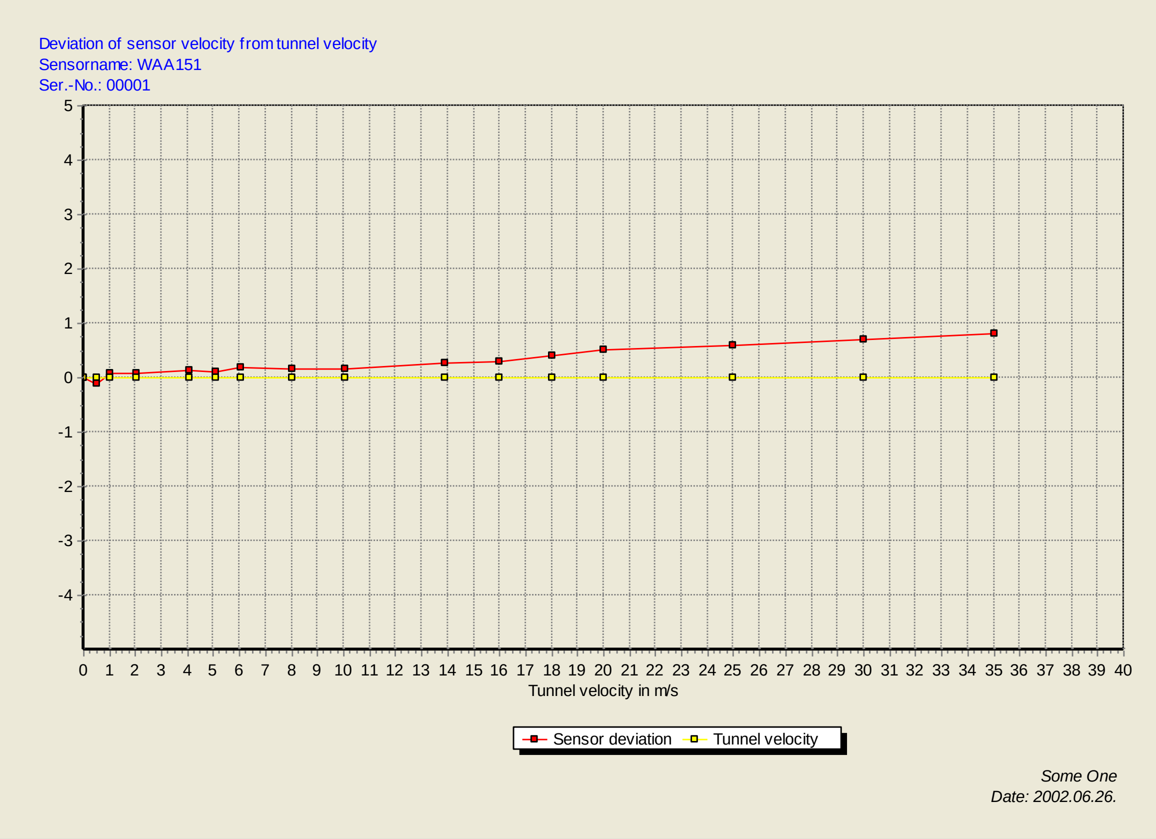
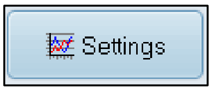


Figure 8: Deviation diagram

Zooming a part of the diagram is possible. To zoom in on a Chart, press the right mouse button at the top left hand corner of the area you wish to zoom in on and, maintaining the mouse button pressed, drag out the rectangle to the bottom right hand corner of the zoom area. Releasing the mouse button will force the Chart to redraw the area selected.

To undo the zoom, press the left mouse button anywhere on the Chart area and drag up and left with the mouse button depressed. Releasing the button will force the Chart to redraw to the originally defined Chart area.

All diagrams can be directly printed or saved as WMF file.

The shape of the chart lines and points, the colour and the axis scale can be adjusted individually by clicking to the 

button above the chart. These settings will be stored in the measuring protocol too.

If you click to ***save as default*** in the diagram settings dialog, these settings will be taken for all new measuring protocols.

1.2.3 ZERO CALIBRATION OF MICROMANOMETER AND THERMAL ANEMOMETER

For measurement of the wind speed of the tunnel the micromanometer and the thermal anemometer are used. The first device is connected via an USB-cable to the PC and the second device is connected to an analogue input channel of the Combilog. Before starting any measurement a zero calibration must be performed at the micromanometer and the thermal anemometer. That means, both values should display zero on the PC screen.

A zero calibration can be performed via the software. Choose ***Measurement* | *Zero calibration*** from the main menu or type the ***F4*** key. In the following dialog box the actual values of the thermal anemometer and the micromanometer are displayed:

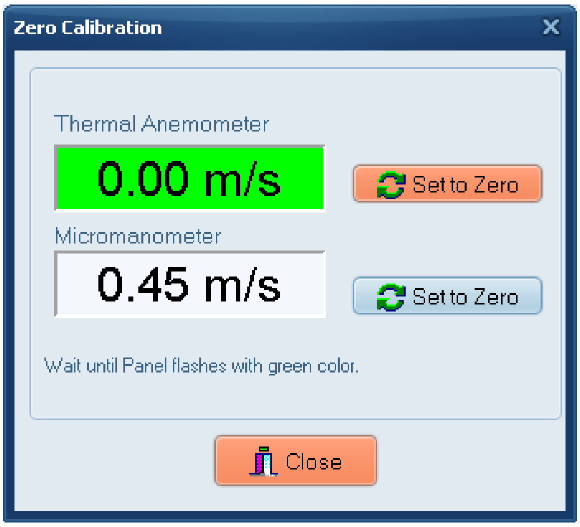


Figure 9: Zero calibration

In case these values are not exactly zero, click to 

. The program will use the displayed value for offset correction for all following measurements.

Note:

The zero calibration for the *Micromanometer* takes about 3 seconds and for the *thermal anemometer* about 1 second. After a zero calibration is done successfully the panel background will be green. So please wait 3 seconds until the panel background will flashes with a green colour after pushing the zero-calibration-button, before starting any measurement.

Note:

After the zero calibration for the *Micromanometer* the displayed value fluctuates about ±0.5 the zero around. This is no failure, because the accuracy of the micromanometer increases with increasing wind speed of Wind tunnel and is acceptable at wind speeds above 1 m/s.

Driving Wind Tunnel velocity manually

For driving the Wind Tunnel velocity manually, perform the following steps:

1. Click into the ***mask*** at the panel of the *Desired speed of Wind tunnel* named **Desired** and insert the desired value or alternatively use the ***regulating bar*** (See Figure 10).
   In case of using the regulating bar:

   - Large steps of 5 m/s resp. 10 kn will be performed by clicking ***into*** the scrollbar.
   - Small steps of 0.05 m/s resp. 0.1 kn will be performed by clicking ***on*** the arrows of the scrollbar.
   - By clicking on the bar and scrolling it, any desired value can be chosen.

2. The *Actual speed of the Wind tunnel* is shown at the panel named **Reference**. The **Synchronized** LED will flash with a green colour, as soon as the Wind Tunnel reaches the desired speed. Check “Display Mean Value” to show the mean value, otherwise the instant value will be displayed.

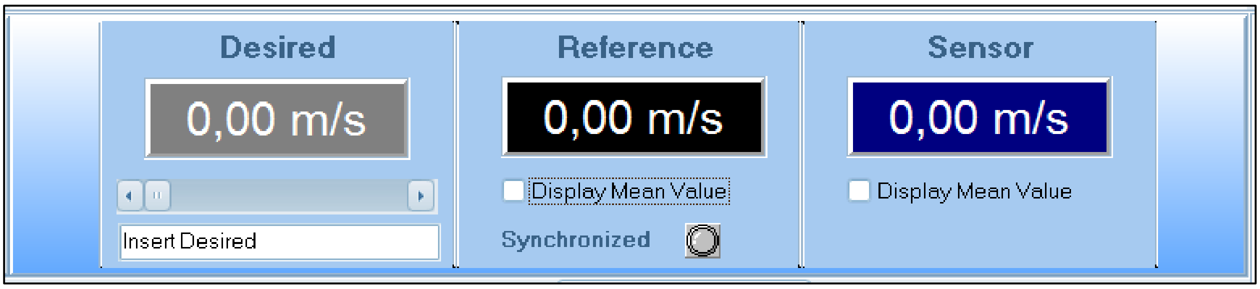


Figure 10: Three Panels displaying Wind speeds

1.2.4 MEASURING OF STARTING THRESHOLD

The starting threshold is defined as the wind velocity the sensor begins to turn. As the output signal of the sensor at this moment is almost Zero, the corresponding wind velocity of the tunnel represents the threshold value.

To measure the starting threshold perform the following steps:

1. Click into the ***mask*** at the panel of the *Desired speed of Wind tunnel* named **Desired** and set *manually* the Desired Speed of Wind Tunnel rated value to Zero. Alternatively use the ***regulating bar*** (See Figure 11).

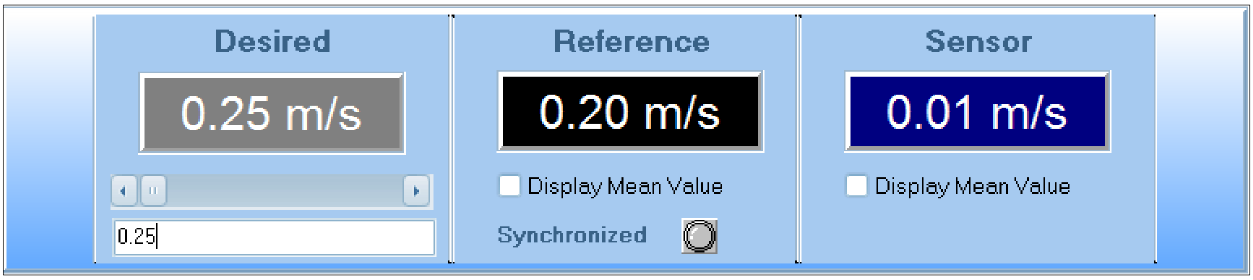


Figure 11: Setting Desired speed of Wind tunnel for measuring a starting Threshold

2. Now the Wind Tunnel must be forced very slowly, during the sensor is watched simultaneously.

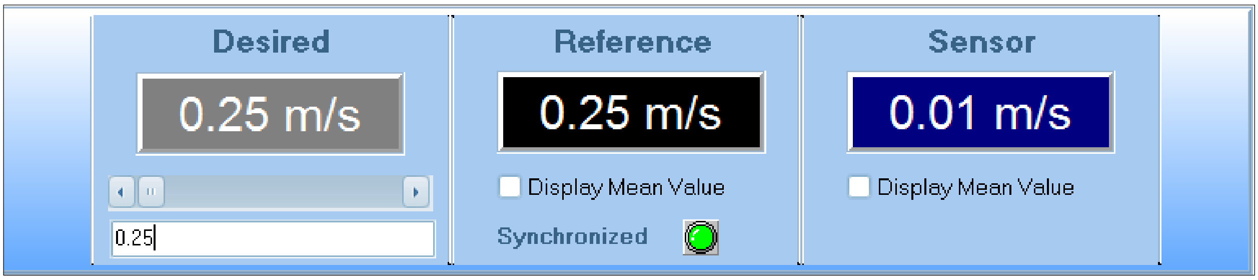


Figure 12: A synchronized Desired and Reference wind speed of Wind tunnel

As the sensor begins to turn, click to  to save this moment as starting threshold into the *measuring protocol*. Instead of clicking with the mouse you can easily press the ***F2*** key or choose ***Measurement | Starting Threshold*** from the main menu.

In the *measuring protocol* a threshold measurement is signed as ***Threshold*** in the column ***Type of measurement***. Since the sensor just begins to output a signal, only the reference value is the threshold value.

1.2.5 NORMAL WIND VELOCITY MEASUREMENT

Measurements of wind velocity can be performed in any order and as many as you like.

1. At the main Window click into the ***mask*** at the panel of the *Desired speed of Wind tunnel* named **Desired** and set *manually* the Desired Speed of Wind Tunnel. Alternatively use the ***regulating bar*** (See Figure 13).

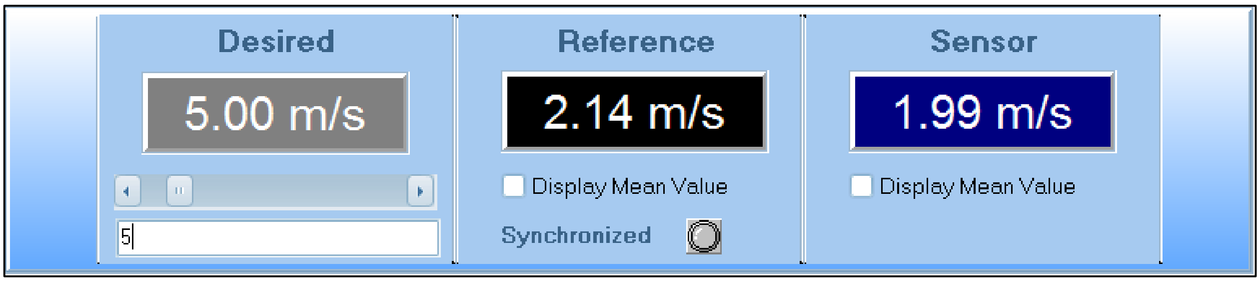


Figure 13: Setting Desired speed of Wind tunnel for normal measurement

2. As the tunnel velocity has stabilized (synchronized), click to  to save the actual sensor velocity into the *measuring protocol*. Instead of clicking with the mouse you can easily press the ***F3*** key or choose ***Measurement | wind velocity*** from the main menu.

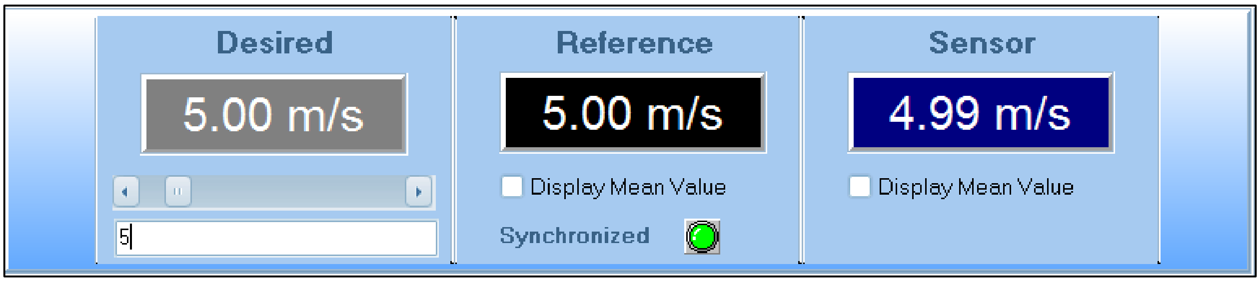


Figure 14: Making manually a normal measurement

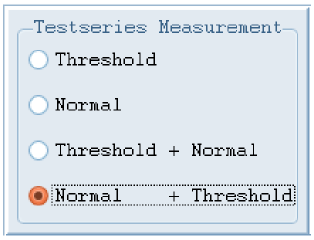
This measurement can be repeated several times.

In the *measuring protocol* a velocity measurement is signed as ***normal*** in the column ***Type of measurement***.

1.2.6 AUTOMATIC MEASUREMENTS WITH TESTSERIES

With predefined Testseries there is the ability given to perform the measuring of the Starting Threshold and multiple Normal Wind Velocities without manually adjusting the Wind Tunnel velocity for each measurement-point.

With predefined Testseries measurements of wind velocity can be performed in any order and as many as you like.



1. At the main Window choose one of the four kinds of Testseries.

   The last two choices are combined Testseries.

2. Click to  or choose ***Measurement | Automatic Measurement*** from the main menu or press easily the ***F5*** key.

3. A dialog box appears, showing the kind of measurement, the desired tunnel velocity, the actual tunnel velocity, the sensor velocity and the delay time.

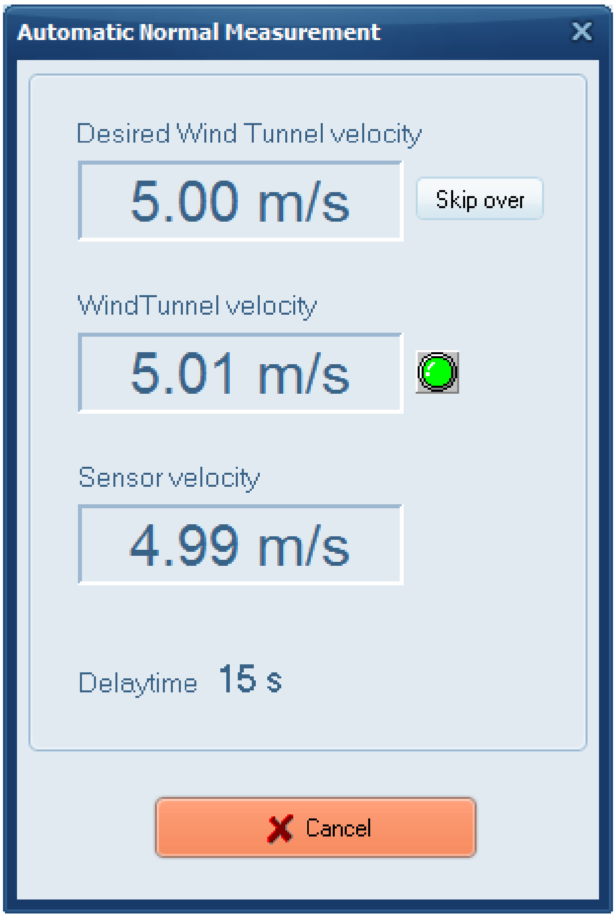


Figure 15: Automatic normal Measurement with Testseries

The LED next to the actual tunnel velocity flushes green, if the desired tunnel velocity is reached.
In case of **Normal Measurement** the delay time counts down until zero. Now a measurement will be performed. Then the measured values are taken. The program takes the next measurement point after storage of the previous point.
The Desired Wind tunnel velocity and the Delay time are taken from the predefined Testseries files, e.g. “normal.WMP”.

In case of **Threshold Measurement** the measurement will be performed, when the shell star begins to turn resp.there is no zero-signal at the Sensor output. Sometimes this Threshold Measurement must be repeated. This can alternatively happen by the way described in chapter 1.2.4 .

In the *measuring protocol* a velocity measurement is signed as ***normal*** in the column ***Type of measurement*** and a threshold measurement is signed as ***Threshold***. The sensor velocity will be set to the real output value of the sensor.

The Skip over button offers the possibility to skip over the actual measurement-point. 

1.2.7 PREDEFINING TESTSERIES

There are two kinds of Test series:

a. **Threshold Measurement Testseries**: This are test series for measurement of the *threshold* wind speed of a sensor.

b. **Normal Measurement Testseries**: This are test series for measurement of the wind speed of a sensor.

Both kinds of test series can be predefined by the following steps:

1. Choose ***Measurement* | *Predefine Testseries*** from the main menu or easily press the ***F6*** key.
2. Click on first row and insert the desired **Wind speed of the Tunnel** and the desired **Holdtime**, which is the time in seconds holding the real wind speed of the Wind Tunnel after synchronization.
3. Use the ***Add***, ***Delete*** and ***Delete all*** buttons for adding or deleting new measurement points in the test series.

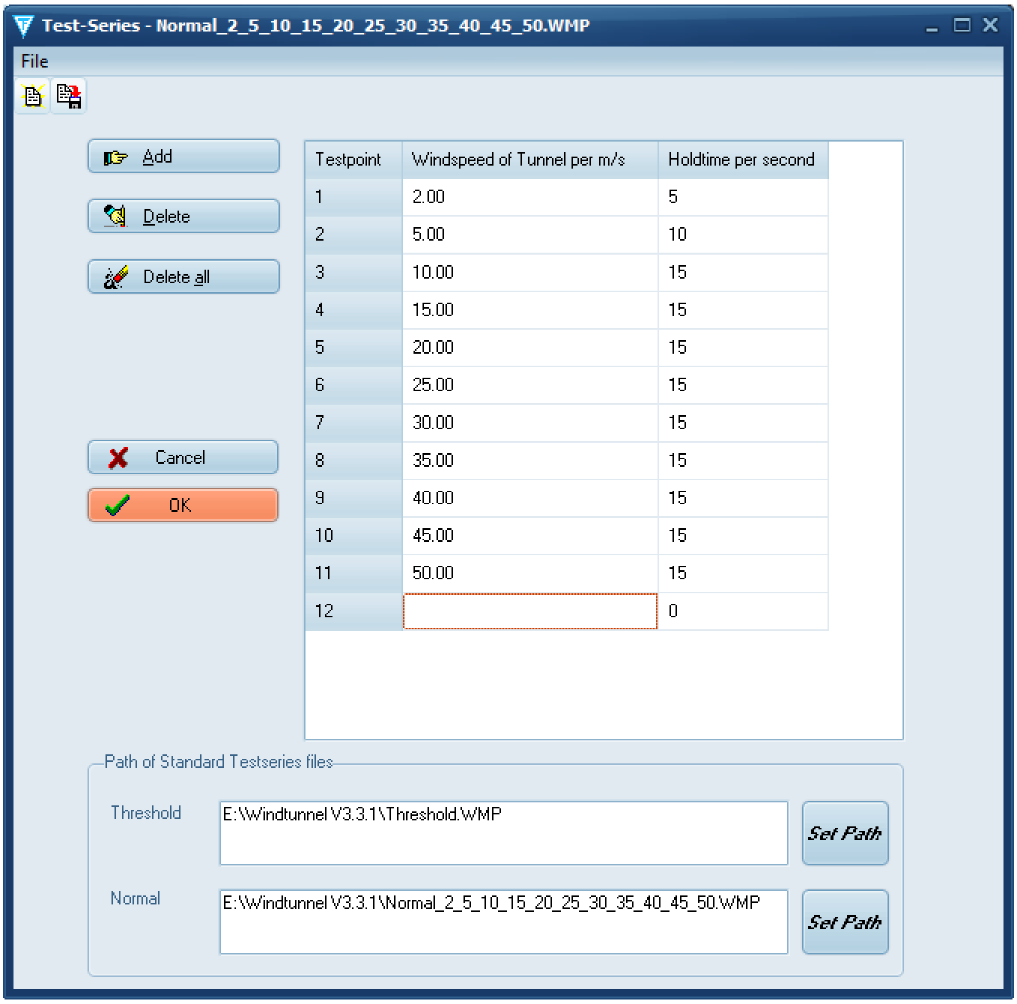


Figure 16: Predefining Test Series

4. Save the file by choosing **save** or **save as** from the menu or by clicking on the button .

5. Choose the *path of the test series files* for ***Threshold Measurement*** and ***Normal Measurement*** by clicking on each button .

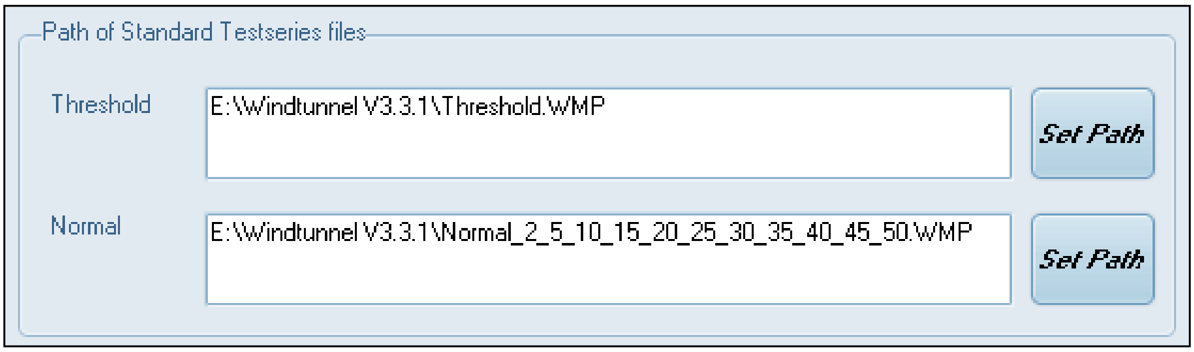


Figure 17: Path of Standard Testseries files

This are now the Testseries described in chapter 1.2.6 !

2 ADDITIONAL PROGRAM SETTINGS

The basic settings of the program can be set by ***Setup | Program settings*** in the following dialog box:

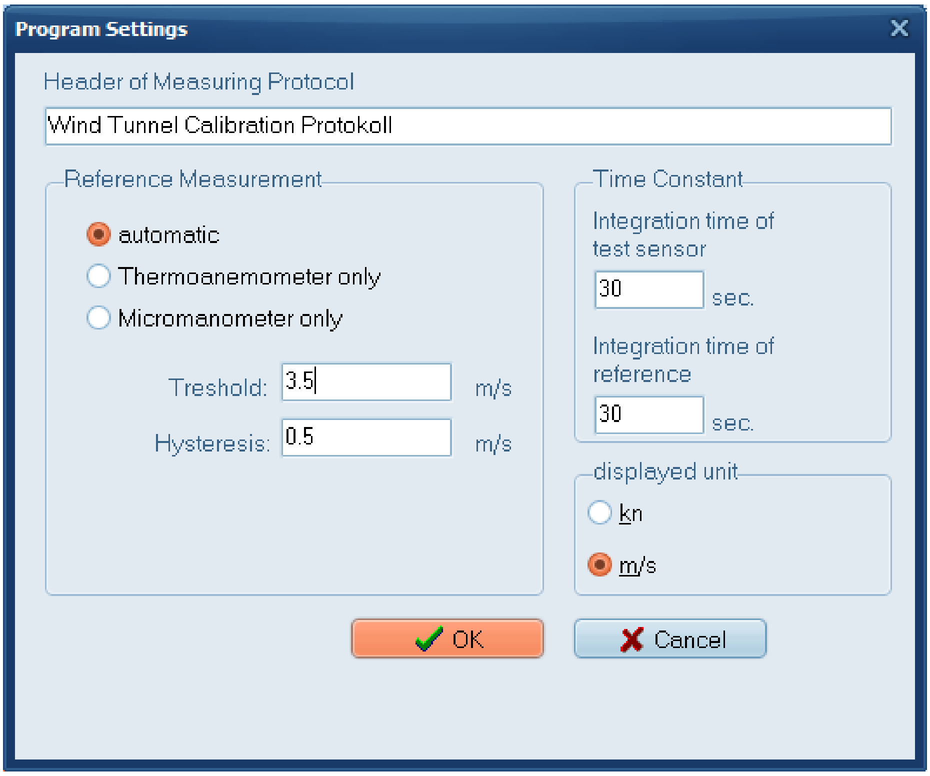


Figure 18: Program settings

The ***Header of Measuring protocol*** is printed in the first line of the protocol, when it is send to the printer. Any text can be entered.

The ***Reference measurement*** box defines the used reference device to measure the tunnel velocity.
In case of ***automatic*** selection the ***Threshold*** value defines, at which velocity level the thermal anemometer resp. the micromanometer is used to measure the tunnel velocity. As the thermal anemometer has a higher accuracy at lower wind speed values, this device will be used in case the tunnel velocity is lower than the specified level. To avoid permanent switching between thermal anemometer and micromanometer around the level speed, a ***hysteresis*** is defined. This is the value the velocity value must fall below the threshold value to switch to the thermal anemometer again. To switch between the two references only the wind speed value of the thermal anemometer is crucial. The *factory setting* are presetted for the ***Threshold*** 3.5 m/s and for the ***Hysteresis*** 0.3 m/s.

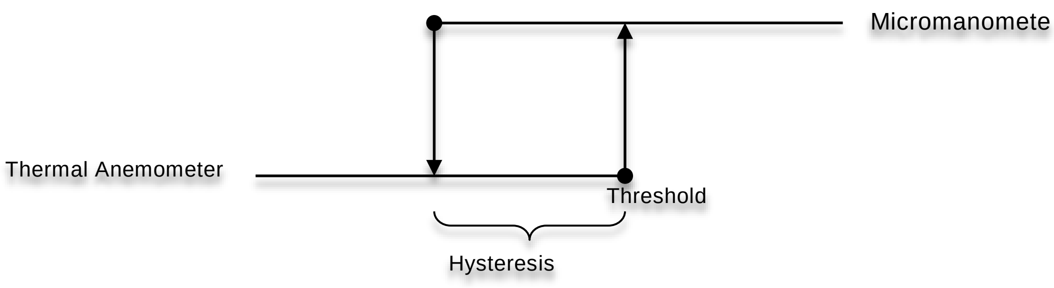


Figure 19: Hysteresis of Reference measurement

The ***displayed unit*** can be selected between ***kn*** and ***m/s***. If a measuring protocol with data is opened, all measuring values will be recalculated and displayed in the corresponding unit.

Furthermore the communication port can be changed by ***Setup | Communication*** in case it should be necessary. Note, that this function is only available in offline mode.

3 DYNAMIC DATA

The history of wind speeds data is named "*Dynamic data*".
The ***Dynamic data*** is associated to the following values (see figure 20):

- **Desired**:
  The *desired* wind speed value of the Wind tunnel
  which is shown at the left grey panel.

- **Reference**:
  The *instant* wind speed value of the *Reference*
  which is either taken from the Thermal anemometer or the Micromanometer.

  If the checkbox named "*Display Mean Value*" is **unchecked**, then this value is shown at the middle black panel.

- **Reference mean**:
  This is the mean value of the *Reference*.
  Its Time constant can be set at the integration time of Reference (see figure 18).

  If the checkbox named "*Display Mean Value*" is **checked**, then this value is shown at the middle black panel.

- **Sensor**:
  The *instant* wind speed value of the *Sensor*.

  If the checkbox named "*Display Mean Value*" is **unchecked**, then this value is shown at the right blue panel.

- **Sensor mean**:
  This is the mean value of the *Sensor*.
  Its Time constant can be set at the integration time of test sensor (see figure 18).

  If the checkbox named "*Display Mean Value*" is **checked**, then this value is shown at the right blue panel.

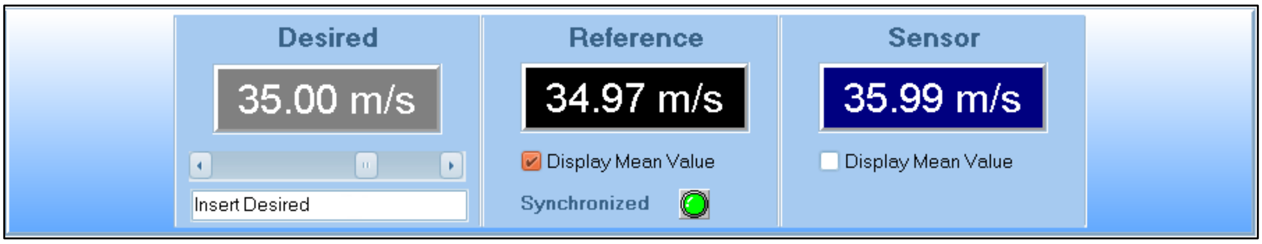


Figure 20: 3 Panels at the main window showing wind speed values

3.1 DYNAMIC DATA ANIMATION

The following figure shows the main window with index tab which is named "*Dynamic Data Animation*".

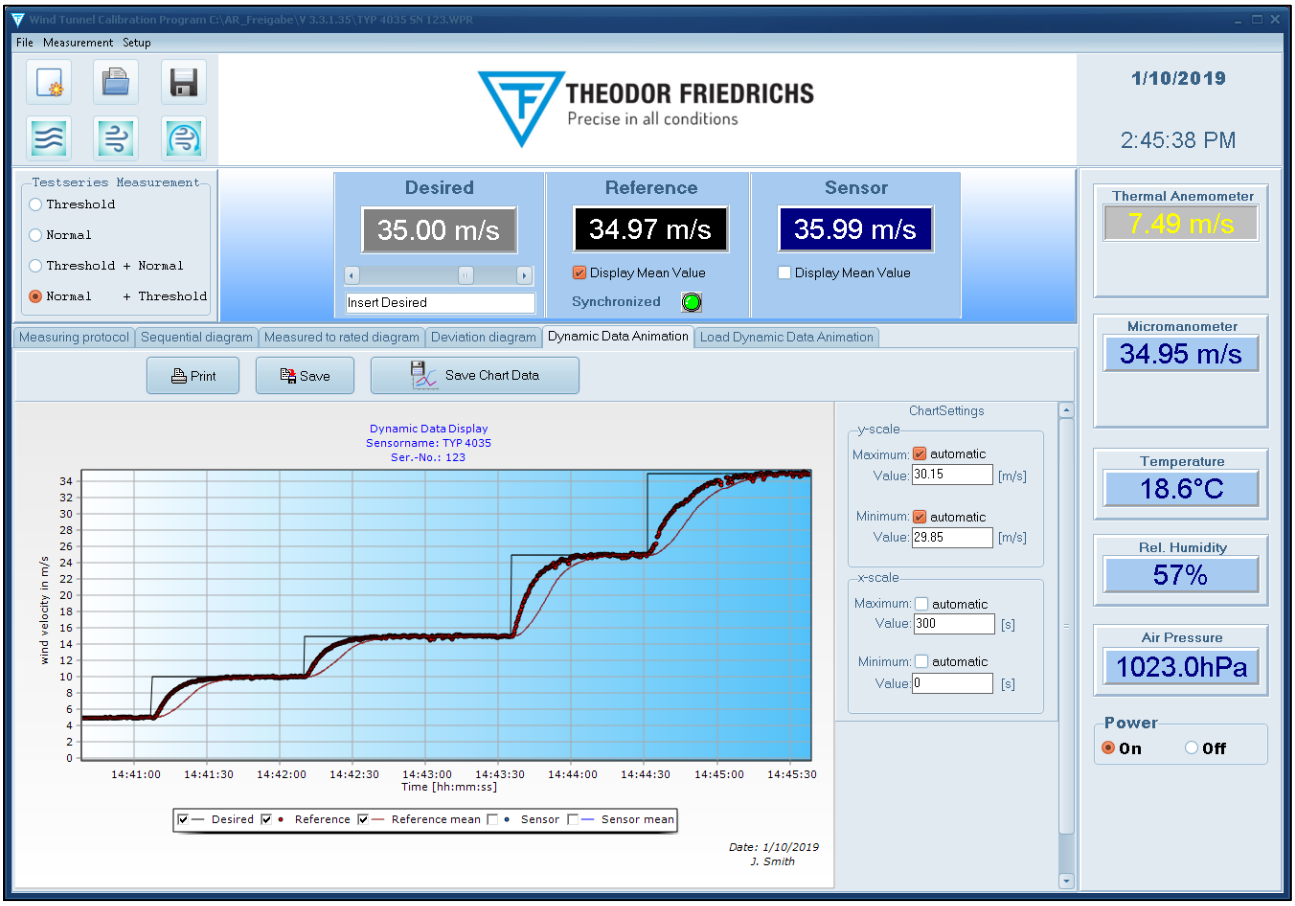


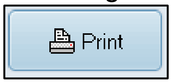
Figure 21: Main window with animation of the Dynamic data

The Dynamic Data Animation gives the possibility to show a history of wind speeds. It's a diagram with one x-axis (horizontal axis) and one y-axis (vertical axis). The ***x-axis*** indicates the ***time***, when the measuring happened. This time format is displayed as hours, minutes and seconds, e.g.: **08:53:00** has the following time components:

hh	= 08	, that are 8 hours,
mm	= 53	, that are 53 minutes and
ss	= 00	, that are 0 seconds.

The *range* of the x-axis can be set as *integer seconds*.
The ***y-axis*** indicates the measured ***wind speed value*** in the predefined units m/s or kn.
The axes range can be scaled by the panel "*Charts settings*" at the right of the diagram.
Every minimum or maximum of the axes can be selected automatically by checkmarking or by typing in the value.

The diagram shows graphs of the *desired wind speed value* (named “***Desired***”), the *wind speed value* measured with a reference sensor (named “***Reference***”), the mean value of the *Reference* (named “***Reference mean***”), the *wind speed value* measured with a Sensors under test (named “***Sensor***”) and the mean value of the *Sensor* (named “***Sensor mean***”). It is also possible to checkmark the graphs, that you want to see.
E.g.: in figure 21 only the check marked graphs of “Desired”, “Reference” and “Reference mean” are shown.

The diagram of Dynamic Data Animation can be send to a printer by clicking the 

button and by clicking the 

button its picture can be saved to a WMF file (Windows Metafile).

Hint:
Whenever a new measuring protocol is started, the diagram will be cleared, that is the history is erased!

Before starting a *new protocol*, the whole ***chart data*** can be saved as ***TEE files*** by clicking the 

button.

The ***chart data*** can be loaded at a later time as described in the following subchapter 3.2.

3.2 LOAD DYNAMIC DATA ANIMATION

The following figure shows the main window with index tab which is named "*Load Dynamic Data Animation*":

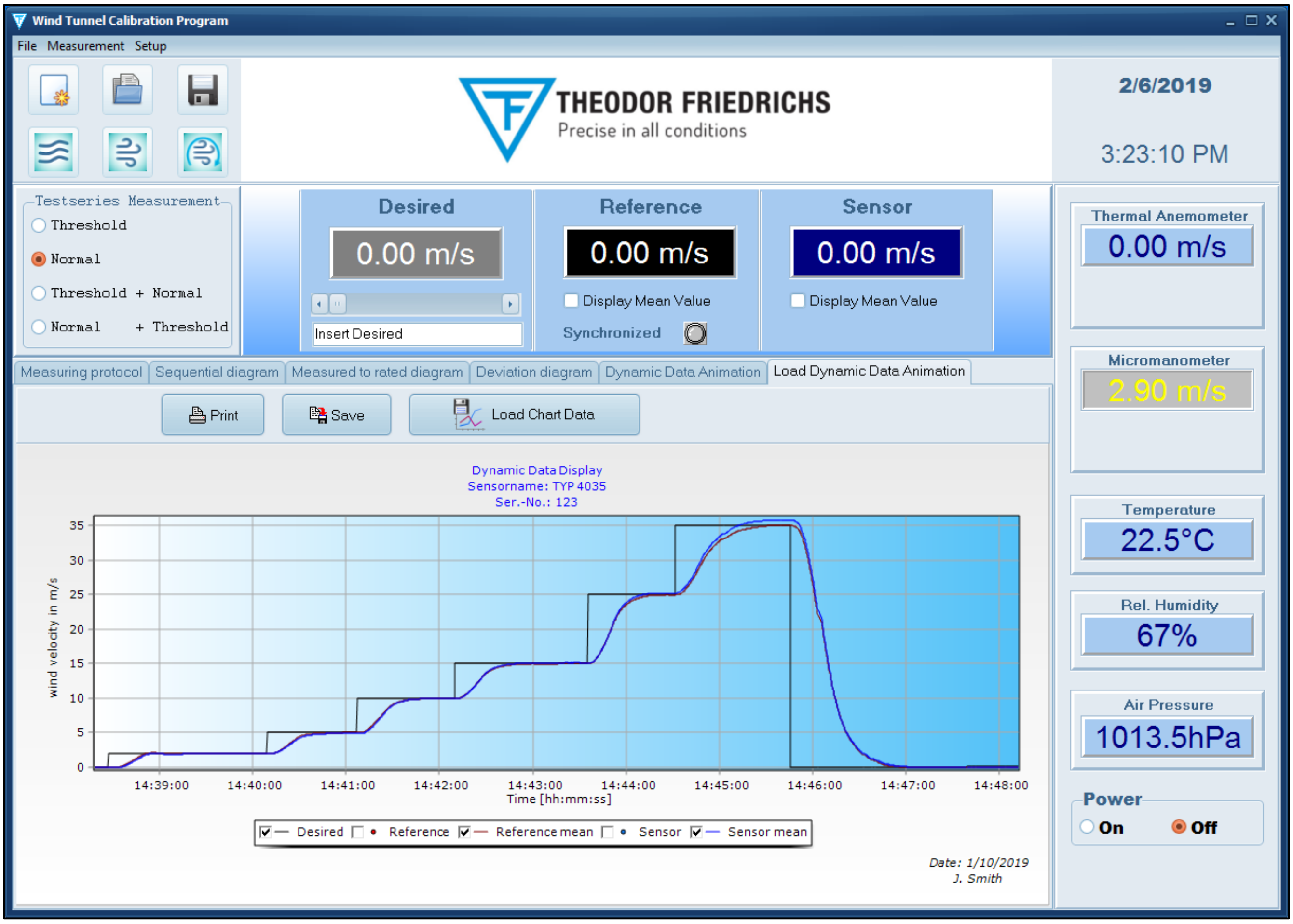
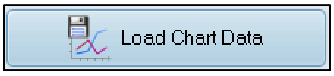


Figure 22: Main window with loaded chart data

A ***chart data*** which was saved as a ***TEE file*** can be loaded by clicking the



button. Some protocol information, that are *Sensor name*, *Serial number*, *Operator* and *Date* of operation is shown in the chart (see figure 22).
The diagrams graphs can be selected by checkmarking for showing.
The diagram of Dynamic Data Animation can be send to a printer by clicking the

button and by clicking the

button its picture can be saved to a WMF file (Windows Metafile).

3.2.1 ZOOM

If required, any range of the curves can be *zoomed*. *Zooming* is activated by drawing a rectangle around the chart area of interest with pressed left mouse button, beginning from top / left down to bottom down. Dragging in the opposite direction resets axis scales.
Zooming can be repeated several times. The zoomed range can be scrolled in any direction by moving the mouse with pressed right mouse button.

E.g.: There is a chart shown in figure 23 which is a zoomed area of the chart of figure 22.

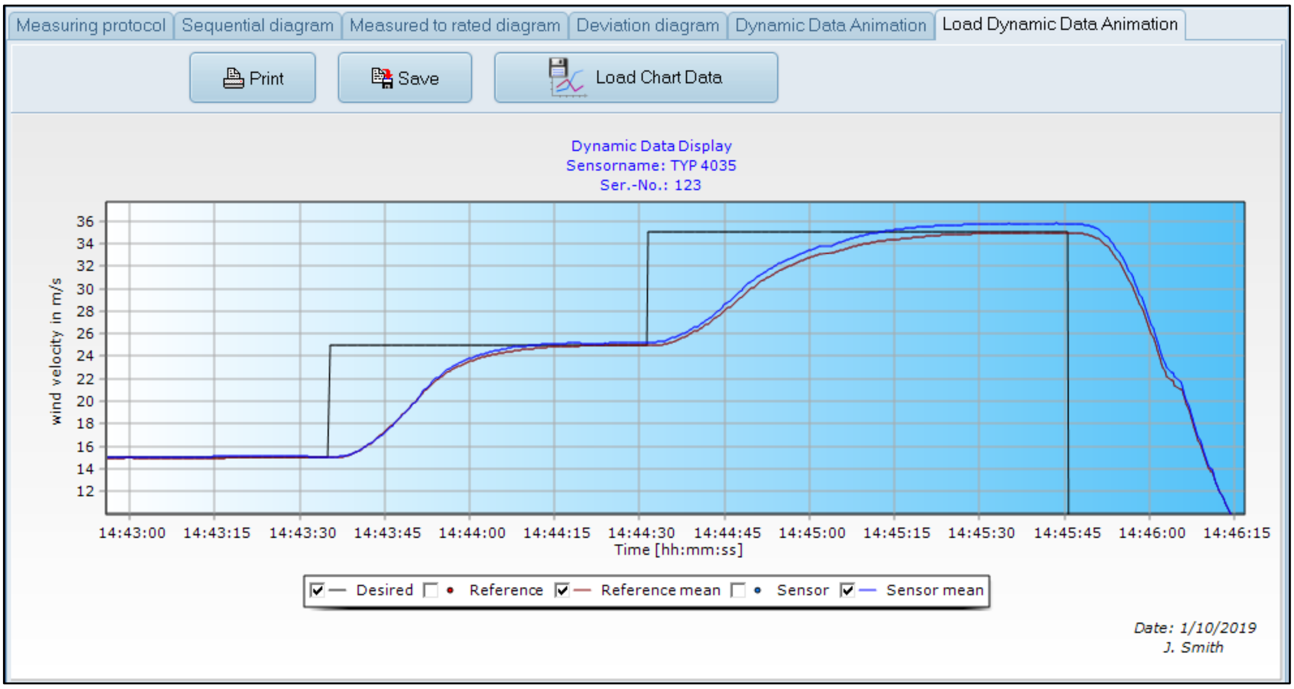


Figure 23: Zoomed area of loaded chart data